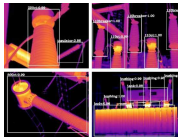
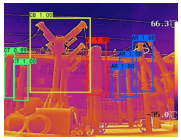
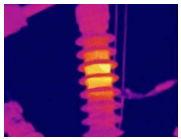
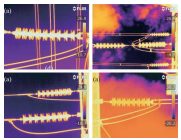
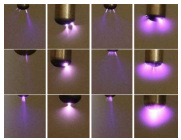
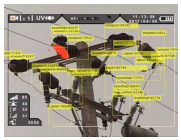
Task Description	Data				Model			
	Sample	Size	Annotations	Source	Algorithm	Architecture	Performance	Ref
Substations Assets Health Condition Diagnostic		12,683 Images 640 x 480 x 1	16 Classes 18,146 B. Box Num.	Not Available	Supervised DL	Retina-Net	mAP50 = 85.3%	[252]
					Supervised DL	Modified YOLOv4	mAP50 = 86.4%	[252]
Substations Assets Detection		1,263 Images 600 x 600 x 1	5 Classes Unspecified B. Box Num.	Not Available	Supervised DL	YOLOv5 CSPdarknet53	mAP50 = 91.85%	[253]
					Supervised DL	Faster R-CNN VGG16	mAP50 = 92.78%	[253]
HV Insulators Temperature Measurement		1,200 Images 1024 x 1024 x 1	1 Classes Unspecified B. Box Num.	Not Available	Supervised DL	Mask R-CNN ResNet101-FPN	mAP50 = 77%	[254]
HV Insulators Segmentation		2,540 Images 640 x 480 x 1	1 Classes Unspecified B. Box Num.	Not Available	Supervised DL	BSIS ResNet101-FPN	mAP50 = 95.19%	[255]
Corona Detection and Discharge Mode Classification		741 Images 500 x 500 x 3	4 Classes Unspecified B. Box Num.	Open Available [256]	Supervised DL	YOLOv8m	mAP50 = 99.3%	[257]
					Supervised DL	ShuffleNet	F1 = 100%	[257]
Corona Detection and Power Equipment Classification		462 Images 4 Channels Unspecified Resolution	7 Classes 4,486 B. Box Num.	Not Available	Supervised DL	Faster R-CNN	mAP = 78%	[51]
					Supervised DL	Optimized Faster R-CNN	mAP = 82%	[258]
Substations Assets Instance Segmentation		1,660 Images 640 x 480 x 3 4672 x 3504 x 3	15 Classes 50,705 B. Box Num.	Data + Code Available [259], [260]	Weak-Supervised DL	DeepLabV3+ GrabCut SDI	IoU = 62.9%	[261]